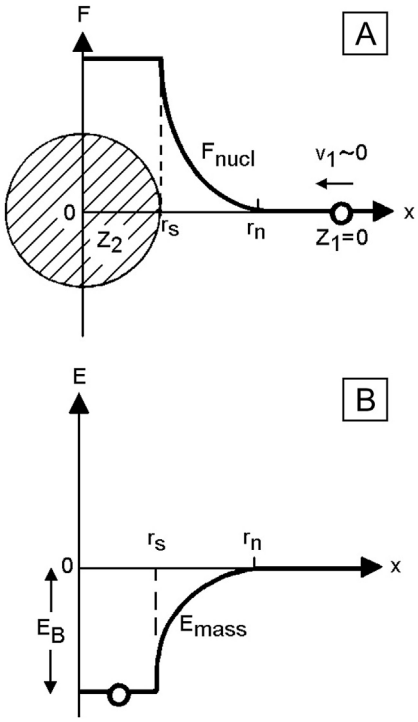


Figure 6.1 The approach along the x-axis of a neutron (charge $Z_1 = 0$, velocity $v_1 \approx 0$) towards a nucleus (charge Z_2; Fig. A) leading to the neutron being trapped at a certain energy level, E_B (Fig. B). F_{nucl} is the nuclear force extending from the surface of the "solid core" at r_s to the distance r_0. E_B is the nuclear binding energy in the potential well.

protons as for neutrons. Figure 6.2 shows the energy levels of the protons and of the neutrons in the nucleus of $^{116}_{50}$Sn.

6.2. ROTATIONAL ENERGY AND ANGULAR MOMENTUM

It is an intriguing fact that nucleons in nuclei, electrons in atoms, as well as large cosmic objects such as solar systems and even galaxies, are more dominated by rotational than by linear motion, although in our daily life the latter seems to be a more common phenomenon. In rotation there is a balance between two forces: the centrifugal force of inertia, which tries to move a body away from a center point, and an attractive force (gravitational, electrostatic, etc.), which opposes the separation.

In the preceding section we assumed that the nucleus existed in some kind of a potential well. One may further assume that a nucleon moves around in this well in a way not too different from the way the electron moves around the atomic nucleus, i.e. with an oscillation between kinetic and potential energy. With some hypothesis about the shape

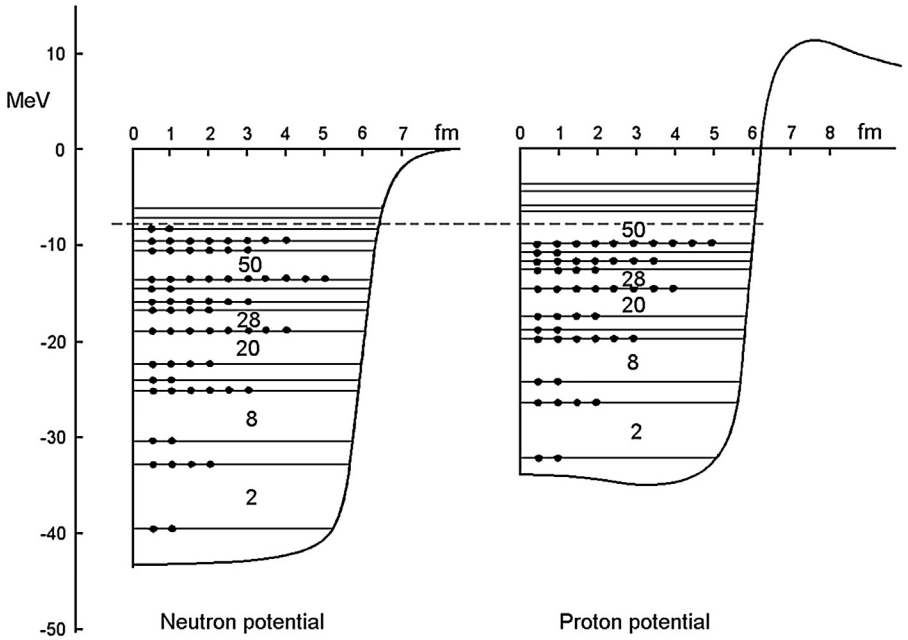


Figure 6.2 The ^{116}Sn neutron and proton shell structure in the potential well. (According to S. G. Nilsson.)

of the nuclear potential well we can apply the *Schrödinger wave equation* to the nucleus. Without being concerned at this point with the consequences of assuming different shapes we can conclude that the solution of the wave equation allows only certain energy states. These energy states are defined by two quantum numbers: the *principal quantum number n*, which is related to the total energy of the system, and the *azimuthal* (or *radial*) *quantum number l*, which is related to the rotational movement of the nucleus.

6.2.1. Rotational (mechanical) energy

If a mass m circles in an orbit of radius r at a constant angular velocity ω (rad s^{-1}), the tangential velocity at this radius is

$$v_r = \omega r \tag{6.1}$$

The kinetic energy in linear motion is $E_{kin} = \frac{1}{2}mv^2$, and we therefore obtain for the (kinetic) rotational energy

$$E_{rot} = \frac{1}{2}\, m\omega^2 r^2 \tag{6.2}$$

The angular velocity is often expressed as the frequency of rotation, $\upsilon_r = \omega/2\pi$ s^{-1}. Equation (6.2) can also be written

$$E_{rot} = \frac{1}{2}\, I_{rot}\omega^2 \tag{6.3}$$

Where

$$I_{\text{rot}} = \Sigma m_i r_i^2 \tag{6.4}$$

is the rotational *moment of inertia*. We consider as the rotating body a system of i particles of masses m_i, each individual particle at a distance r_i from the axis of rotation; then (6.3) is valid for any rotating body. In nuclear science we primarily have to consider two kinds of rotation: the *intrinsic rotation* (or *spin*) of a body around its own axis (e.g. the rotation of the earth every 24 h), and the *orbital rotation* of an object around a central point (e.g. rotation of the earth around the sun every 365 days). Eqn (6.3) is valid for both cases, but (6.2) only for the orbital rotation of a particle of small dimensions compared to the orbital radius, in which case $I_{\text{rot}} = mr^2$. For a spherical homogeneous spinning body (e.g. the earth's intrinsic rotation) of external radius r_{ex}, (6.3) must be used, where $I_{\text{rot}} = 2mr_{ex}^2/5$.

6.2.2. Angular momentum

Like linear motion, rotation is associated with a momentum, called the *angular momentum*. For the orbital rotation the *orbital angular momentum* (p_l) is

$$p_l = mv_{\text{r}}r \tag{6.5}$$

while for spin the spin angular momentum (p_s) is

$$p_s = \omega I_{\text{rot}} \tag{6.6}$$

Angular momentum is a *vector quantity*, which means that it has always a certain orientation in space, depending on the direction of rotation. For the rotation indicated in Figure 6.3(a), the vector can only point upwards. It would not help to turn the picture

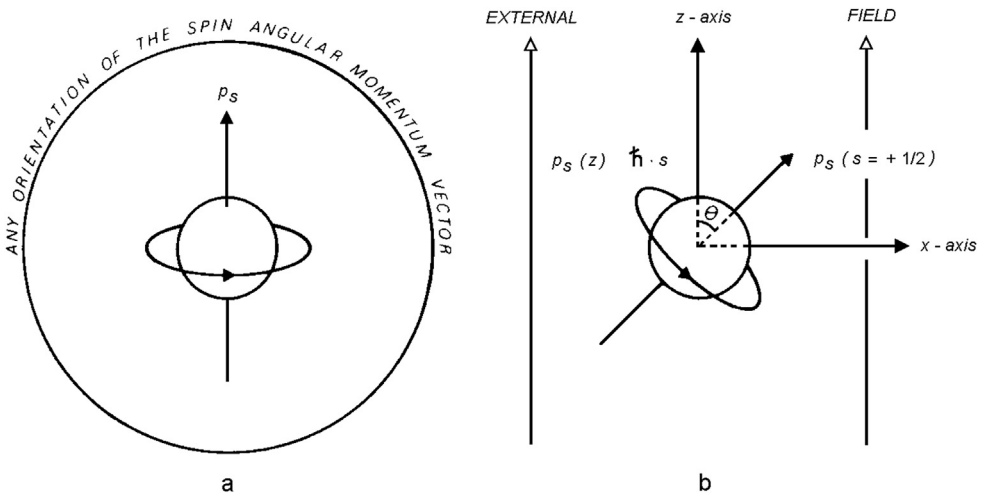


Figure 6.3 A particle spinning in (a) field-free space and (b) in an external field along the *z*-axis.

upside down because the coupling between rotational direction and its vector remains the same, as students south of the equator will agree.

Quantum mechanics prescribes that the spin angular momentum of electrons, protons, and neutrons must have the magnitude

$$p_s = \hbar[s(s+1)]^{1/2} \tag{6.7}$$

here s is the *spin quantum number*. For a *single* particle (electron or nucleon) the spin s is always ½. In addition to spin, the three atomic particles can have orbital movements. Again quantum mechanics prescribes that the magnitude of the orbital angular momentum of these particles

$$p_l = \hbar[l(l+1)]^{1/2} \tag{6.8}$$

We shall refer to l as the *orbital* (*angular momentum*) *quantum number*. Only certain values are permitted for l, related to the main quantum number n:

for electrons: $0 \leq l < n-1$
for nucleons: $0 \leq l$

For nucleons but not for electrons l may (and often does) exceed n.

6.2.3. Coupling of spin and orbital angular moments

A rotating charge gives rise to a *magnetic moment* μ_s. The rotating electron and proton can therefore be considered as tiny magnets. Because of the internal charge distribution of the neutron it also acts as a small magnet. In the absence of any external magnetic field these magnets point in any direction in space (Fig. 6.3(a)), but in the presence of an external field they are oriented in certain directions determined by quantum mechanical rules. This is indicated by the angle θ in Figure 6.3(b), when we have the spinning particle in the center of a coordinate system. The quantum mechanical rule is that the only values allowed for the projections of spin angular momentum $p_s(z)$ on the field axes are:

$$p_s(z) = \hbar m_s \tag{6.9}$$

For composite systems, like an electron in an atom or a nucleon in a nucleus, the (*magnetic*) *spin quantum number* m_s may have two values, +½ or −½, because the spin vector has two possible orientations (up or down) with regard to the orbital angular momentum.

The orbital movement of an electron in an atom, or of a proton in the nucleus, gives rise to another magnetic moment (μ_l) which also interacts with external fields. Again quantum mechanics prescribes how the orbital plane may be oriented in relation to such a field (Fig. 6.4(a)). The orbital angular momentum vector l can assume only such directions that its projection on the field axes, $p_l(z)$, has the values

$$p_l(z) = \hbar m_l \tag{6.10}$$

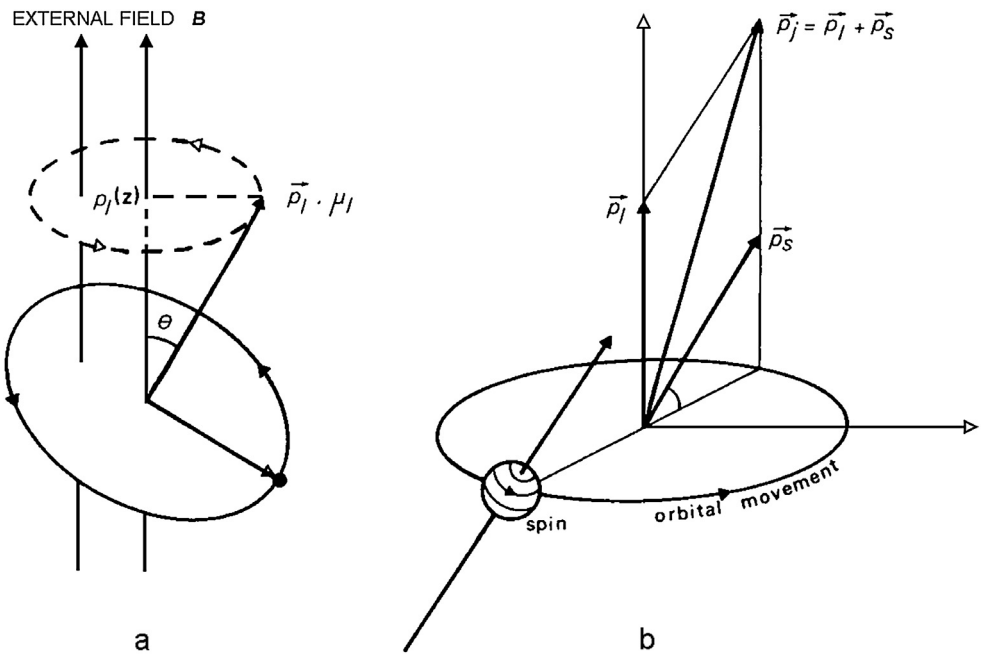


Figure 6.4 (a) Angular momentum caused by an orbiting particle and permitted value projected on the external field axis. The vector p_l precesses around the field axis as indicated by the dashed circle (ellipse in the drawing). (b) The l–s coupling of orbital angular momentum and spin leads to a resultant angular momentum p_j.

m_l is referred to as the *magnetic orbital quantum numbers;* m_l can have all integer values between $-l$ and $+l$.

For a single-particle, nucleon or electron, the orbital and spin angular moments add vectorially to form a *resultant* vector (Fig. 6.4(b)),

$$\overrightarrow{p}_j = \overrightarrow{p}_l + \overrightarrow{p}_s \tag{6.11}$$

$\overrightarrow{p}_j$ will orient itself towards an external field so that only the projections

$$p_j(z) = \hbar m_j \tag{6.12}$$

are obtained on the field axes, m_j is the *total magnetic angular momentum quantum number*; it can have all integer values between $-j$ and $+j$. The magnitude of p_j is

$$p_j = \hbar[j(j+1)]^{1/2} \tag{6.13}$$

where

$$j = l \pm s \tag{6.14}$$

Here j is the *total* (*resultant*) *quantum number of the particle.*

6.2.4. Magnetic moments

a. Single particles. Dirac showed in 1928 that the spin magnetic moment of the electron is:

$$\mu_s(\text{electron}) = \hbar\mathbf{e}/2\mathbf{m}_\text{e} = \mathbf{B}_\text{e} \tag{6.15}$$

where $\mathbf{B}_\text{e} = 9.273 \times 10^{-24}\,\text{JT}^{-1}$ (joule per tesla). This value is referred to as one *Bohr magneton* (B.m.). For the proton one would expect the spin magnetic moment μ_s to be

$$\mathbf{B}_\text{e} \cdot \mathbf{m}_\text{e}/\mathbf{m}_\text{p} = \mathbf{B}_\text{n} \tag{6.16}$$

$\mathbf{B}_\text{n}$ (n for nucleon) has the value $5.051 \times 10^{-27}\,\text{JT}^{-1}$ and is referred to as one *nuclear magneton* (n.m.). However, measurements show that

$$\mu_s\big(\text{proton}\big) = 14.1 \times 10^{-27}\,\text{JT}^{-1} = 2.793\ \text{n.m.} \tag{6.17}$$

The reason for the higher value is found in the uneven charge distribution within the proton.

Recall that the neutron also has an uneven charge distribution. As pointed out earlier, this gives rise to a neutron spin magnetic moment

$$\mu_s(\text{neutron}) = -1.913\ \text{n.m.} \tag{6.18}$$

The magnetic moment μ_l, caused by a charge q in *circular orbit*, can be calculated from classical physics: the current caused in the orbit, qv_r, times the area encircled, πr^2. Therefore

$$\mu_l = qv_\text{r}r/2 \tag{6.19}$$

Dividing (6.19) by p_l according to (6.5), gives

$$\gamma = \mu_l/p_l = q/2m \tag{6.20}$$

This ratio γ is called the *gyromagnetic ratio.* If γ and p_l are known, μ_l may be calculated. Equation (6.20) is valid for electrons ($\mathbf{e}$, $\mathbf{m}_\text{e}$), but for protons only the left part. Because of the quantization of $p_l(z)$, (6.12), the component of the orbital magnetic moment in the field direction (Fig. 6.4(a)) is also quantized:

$$\mu_l(\text{electron}) = m_l\mathbf{B}_\text{e} \tag{6.21}$$

Such a simple approach is not possible for a nucleon in a nucleus, because no nucleon moves completely independent of the other nucleons, nor is the orbital path always circular.

b. Atoms and nuclei. In an atom with many electrons, the spin and angular moments of the electrons couple vectorially and separately to form of resultant quantum numbers (using conventional symbolism)

$$S = \Sigma\overrightarrow{s}_i \tag{6.22}$$

$$L = \Sigma\overrightarrow{l}_i \tag{6.23}$$

which couple to form the total angular momentum (or internal) quantum number of the atom

$$\overrightarrow{J} = \overrightarrow{L} + \overrightarrow{S} \tag{6.24}$$

(see Table 6.2). Equations (6.22) to (6.24) are referred to as *Russell-Saunders coupling*. For the atom as a whole, the magnetic moment is

$$\mu(\text{atom}) = g_j \, \mathbf{e} \, p_j/2 \, \mathbf{m}_\text{e} = g_j \, \mathbf{B}_\text{e} \, m_j \tag{6.25}$$

where g_j is the Landé factor and $+J \leq m_j \leq -J$. The Landé factor accounts for the effect of mutual screening of the electrons.

The *nuclear magnetic moment* depends on the spin and angular moments of the neutrons and protons. For the nucleus it is given by

$$\mu(\text{nucleus}) = g_I \, \mathbf{e} \, p_I/2 \, \mathbf{m}_\text{p} \tag{6.26}$$

where g_I is the nuclear g-factor and p_I, the magnitude of the nuclear spin angular moment. Because this moment can have only the projections $\hbar m_I$ on the axes of a magnetic field where m_I is the nuclear magnetic angular momentum quantum number ($-I \leq m_I \leq I$), we may write this equation

$$\mu(\text{nucleus}) = g_I(\mathbf{e}\,\hbar/2\, m_p) m_I = g_I \, \mathbf{B}_\text{n} \, m_I \tag{6.27}$$

I is the *total nuclear spin*. We shall see in the next section how I can be determined.

In Table 6.1 we have summarized the most important properties of the atomic constituents, and in Table 6.2 their modes of interaction. In Table 6.3 some spin and magnetic moments are given for stable and radioactive nuclei.

6.2.5. Precession

Before going into the details of nuclear structure, there is one more property of the nucleon which must be considered. Both types of angular momenta, p_s and p_l, as well as their corresponding magnetic moments, are vector quantities. Quantum mechanics forbids $\overrightarrow{p}$ (and consequently μ) to be exactly parallel with an external field. At the same time the external field tries to pull the vector so that the plane of rotation becomes perpendicular to the field lines. The potential magnetic energy is

$$E_\text{magn} = \overrightarrow{B} \cdot \overrightarrow{\mu} = B \cdot \mu \cos\theta \tag{6.28}$$

(see Figs. 6.3 and 6.4) where B is the magnetic field strength. However, because of the inertia of the particle this does not occur until the particle has moved somewhat along its orbit, with the consequence that the vector starts to rotate around the field axes as shown in Figure 6.4(a). The angular momentum vector therefore precesses around the field axes, like a gyroscope axis. We may rewrite (6.20) as

$$\gamma = \overrightarrow{\mu}/\overrightarrow{p}_\text{rot} \tag{6.29}$$

Table 6.2 Summary of atomic and nuclear properties associated with particle interactions (q.n. = quantum number)

Line property	Electron-electron interaction	Nucleon-nucleon interaction
10 Spin-spin (*s*-*s*) coupling, q.n.	$S = \Sigma s_i$ (strong)	For each nucleon $j = l \pm s$
11 Total spin magnetic moment	$\mu_s = 2(S(S+1))^{1/2}$ B.m.	and $I = \Sigma j_i$
12 Orbit-orbit (*l*-*l*) coupling, q.n.	$L = \Sigma l_i$ (strong)	For ee nuclei: groundstate $l = 0$
13 Total orbital magnetic moment	$\mu_L = 2(L(L+1))^{1/2}$ B.m.	For eo, oe nuclei: groundstate $l = j$
14 Spin-orbit coupling, q.n.	$J = S+L$ (weak)	For oo nuclei: groundstate varies
15 Resulting angular momentum	$p_J = \hbar(J(J+1))^{1/2}$ B.m.	$p_I = \hbar(I(I+1))^{1/2}$ n.m.
16 Resulting magnetic moment	$\mu_J = g_J(J(J+1))^{1/2}$ B.m.	$\mu_I = g_I(I(I+1))^{1/2}$ n.m.
17 Transition selection rules	$\Delta L \pm 1$, $\Delta J = 0$ or ± 1	$\Delta_I =$ integer, $\Delta_j > 0$
18 Multipole moment	Electric dipole	Electric or magnetic multipole
Electron-nucleon interactions		
19 Grand atomic angular mom., q.n.	$F = J + I = S + L + I$	
20 Grand atomic angular mom.	$p_F = \hbar(F(F+1))$; field projections $0, \ldots, \hbar F$	
21 External field: none	Hyperfine spectrum (hfs): J levels split due to nuclear spin I	
22 External field: weak ($\leq 10^{-2}$ T)	Hfs F-levels split into $2F+1$ levels (Zeeman effect)	
23 External field: average ($\sim$10 T)	Electron-nucleon q.n. decouple, producing $(2J+1)(2I+1)$ levels	
24 External field: very strong	Electron spin-orbit decouple, producing separate S- and L-levels	

Table 6.3 Spin I, parity (+ even, − odd), nuclear magnetic moment (μ n.m.) and quadrupole moment $\hat{Q}$ (10^{-28} m^2) for some nuclides ^{A}X

Stable nuclides				Radioactive nuclides				
^{A}X	I (±)	μ_I	$\hat{Q}$	^{A}X	I (±)	μ_I	$\hat{Q}$	$t_{½}$
^{1}H	1/2 +	+2.793		^{3}H	1/2 +	+2.979		12.33 y
^{2}H	1 +	+0.857	+0.003	^{14}C	0 +			5730 y
^{10}B	3 +	+1.801	+0.085	^{24}Na	4 +	+1.690		14.959 h
^{11}B	3/2	+2.689	+0.041	^{32}P	1 +	−0.252		14.262 d
^{12}C	0 +			^{36}Cl	2 +	+1.285	−0.018	3.01×10^{5} y
^{13}C	1/2	+0.702		^{45}Ca	7/2	−1.327	+0.046	163.8 d
^{14}N	1 +	+0.404	+0.019	^{55}Fe	3/2			2.73 y
^{16}O	0 +			^{60}Co	5 +	+3.799	+0.44	5.271 y
^{17}O	5/2 +	−1.894	−0.026	^{64}Cu	1 +	−0.217		12.70 h
^{19}F	1/2 +	+2.629		^{95}Zr	5/2 +			64.02 d
^{23}Na	3/2 +	+2.218	+0.101	^{131}I	7/2 +	+2.742	−0.40	8.0207 d
^{31}P	1/2 +	+1.132		^{137}Cs	7/2 +	+2.841	+0.051	30.0 y
^{33}S	3/2 +	+0.644	−0.076	^{140}La	3	+0.730	+0.094	1.678 d
^{39}K	3/2 +	+0.391	+0.049	^{198}Au	2	+0.593	+0.68	2.6952 d
^{59}Co	7/2	+4.627	+0.404	^{232}Th	0 +			1.405×10^{10} y (α)
^{87}Sr	9/2 +	−1.094	+0.335	^{235}U	7/2	−0.38	+4.55	7.038×10^{8} y (α)
^{141}Pr	5/2 +	+4.275	−0.059	^{238}U	0 +		13.9	4.468×10^{9} y (α)
^{197}Au	3/2 +	+0.146	+0.547	^{239}Pu	1/2 +	+0.203		2.411×10^{4} y (α)

This form makes it more obvious why γ is referred to as the gyromagnetic ratio. Equation (6.29) is valid both for angular momentum and spin, but of course the value is different for electrons and protons. The angular velocity ω_γ of the precession is found to be

$$\omega_\gamma = \mu \overrightarrow{B} / \overrightarrow{p}_{\text{rot}} \tag{6.30}$$

By replacing angular velocity with frequency ($\omega = 2\pi\upsilon$)

$$\upsilon_\gamma = \gamma \cdot \overrightarrow{B} / 2\pi \overrightarrow{p}_{\text{rot}} \tag{6.31}$$

or

$$\upsilon_\gamma = \gamma \cdot \overrightarrow{B} / 2\pi \tag{6.32}$$

where υ_γ is the Larmor precession frequency.

6.3. THE SINGLE-PARTICLE SHELL MODEL

6.3.1. Quantum number rules

Let us assume that a nucleon moves around freely in the nuclear potential well, which is spherically symmetric, and that the energy of the nucleon varies between potential and